

SIMATS ENGINEERING

ASSESSMENT TOOL 3: MCQ Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*

Assessment Tool Weightage: 10%

Q.No	Question	Option A	Option B	Option C	Option D	Ans
1	Preprocessing improves feature detection by	A) Increasing size	B) Reducing noise	C) Changing format	D) Increasing brightness	B
2	Without preprocessing, detection becomes	A) Better	B) Same	C) Inconsistent	D) Faster	C
3	Grayscale conversion helps in	A) Increasing color	B) Reducing complexity	C) Increasing resolution	D) Adding noise	B
4	Binary representation is useful for	A) Texture detection	B) Foreground separation	C) Color enhancement	D) Contrast increase	B
5	Image representation affects	A) Pixel size	B) Feature visibility	C) Memory only	D) Edge removal	B
6	Edge detection identifies	A) Noise	B) Intensity change	C) Smooth regions	D) Colors	B
7	Canny is preferred due to	A) Speed	B) Low error	C) Memory	D) Size	B
8	Sobel is sensitive to	A) Color	B) Noise	C) Size	D) Resolution	B
9	Weak edges occur due to	A) High contrast	B) Low gradients	C) High resolution	D) Size	B
10	Edge detection fails when	A) Smooth image	B) Strong gradients	C) Noise dominates	D) High resolution	C
11	Corner detection finds	A) Flat regions	B) Edges	C) Intensity variation	D) Noise	C
12	Harris is sensitive to	A) Rotation	B) Translation	C) Intensity	D) Noise	D
13	Corners are useful because	A) Smooth	B) Distinctive	C) Large	D) Less intensity	B

Q.No	Question	Option A	Option B	Option C	Option D	Ans
14	Few corners due to	A) High contrast	B) Excess smoothing	C) High resolution	D) Sharp edges	B
15	Harris works well in	A) Uniform	B) Textured	C) Flat	D) Blurred	B
16	Hough Transform is for	A) Noise removal	B) Shape detection	C) Compression	D) Color	B
17	It works using	A) Ignoring noise	B) Voting	C) Reducing pixels	D) Grayscale	B
18	Poor Hough result due to	A) Clean image	B) Noise	C) Strong edges	D) High resolution	B
19	Hough maps to	A) Frequency	B) Parameter space	C) Colors	D) Edges	B
20	Preprocessing helps Hough by	A) Adding noise	B) Enhancing edges	C) Reducing resolution	D) Removing features	B
21	SIFT is invariant to	A) Noise	B) Scale & rotation	C) Color	D) Resolution	B
22	Matching fails when	A) Similar images	B) Weak features	C) Low noise	D) High resolution	B
23	SIFT works better due to	A) Color	B) Scale invariance	C) Size	D) Grayscale	B
24	Feature matching is used in	A) Compression	B) Recognition	C) Noise removal	D) Histogram	B
25	Incorrect matches due to	A) Strong features	B) Similar descriptors	C) High contrast	D) High resolution	B
26	PCA is used for	A) Edge detection	B) Dimensionality reduction	C) Noise addition	D) Smoothing	B
27	PCA removes	A) Features	B) Correlation	C) Noise	D) Pixels	B
28	PCA improves	A) Data size	B) Efficiency	C) Noise	D) Pixels	B
29	PCA may cause	A) Data loss	B) Noise	C) Edge loss	D) Color change	A
30	PCA is applied after	A) Segmentation	B) Feature extraction	C) Filtering	D) Thresholding	B
31	Preprocessing + features improve	A) Noise	B) Accuracy	C) Size	D) Color	B
32	Edge + Hough improves	A) Noise	B) Shape detection	C) Color	D) Compression	B
33	Corner detection used for	A) Noise removal	B) Matching	C) Smoothing	D) Thresholding	B
34	Feature extraction fails when	A) Noisy image	B) Clear image	C) Large image	D) Small image	A

Q.No	Question	Option A	Option B	Option C	Option D	Ans
35	Preprocessing improves	A) Noise	B) Features	C) Pixels	D) Edges	B
36	First step if edges fail	A) PCA	B) Preprocessing	C) SIFT	D) Hough	B
37	Scale issue solved by	A) Sobel	B) SIFT	C) PCA	D) Thresholding	B
38	Redundant features handled by	A) Edge	B) PCA	C) Hough	D) Smoothing	B
39	Missing corners fixed by	A) Threshold	B) Size	C) Color	D) Resolution	A
40	False Hough detection fixed by	A) Noise	B) Preprocessing	C) Color	D) Resolution	B
41	Shape detection uses	A) PCA+SIFT	B) Edge+Hough	C) Smooth+PCA	D) Hist+SIFT	B
42	Rotation matching uses	A) Sobel	B) SIFT	C) Threshold	D) PCA	B
43	Feature extraction helps	A) Compression	B) Key points	C) Noise	D) Color	B
44	Harris fails in	A) Textured	B) Uniform	C) Noisy	D) High contrast	B
45	Preprocessing needed for	A) Visualization	B) Feature extraction	C) Compression	D) Storage	B
46	Edge output used in	A) PCA	B) Hough	C) Smoothing	D) Threshold	B
47	PCA transforms to	A) Spatial	B) Frequency	C) Feature space	D) Binary	C
48	SIFT uses	A) Color	B) Gradients	C) Resolution	D) Size	B
49	Matching depends on	A) Size	B) Distinctiveness	C) Color	D) Resolution	B
50	Feature pipeline improves	A) Size	B) Accuracy	C) Noise	D) Compression	B
51	What is the primary purpose of preprocessing in image analysis?	Compression	Enhancement	Preparation for feature extraction	Storage	C
52	Image representation refers to:	Storage format	Mathematical description of image	Image size	Color type	B
53	Which preprocessing step removes noise?	Edge detection	Smoothing	Segmentation	Thresholding	B

Q.No	Question	Option A	Option B	Option C	Option D	Ans
54	Grayscale image is represented using:	1 channel	2 channels	3 channels	4 channels	A
55	Feature extraction is used to:	Store images	Reduce redundancy	Compress data	Enhance brightness	B
56	Image preprocessing improves:	Storage	Quality for analysis	Transmission	Display	B
57	Pixel intensity represents:	Position	Brightness	Shape	Color depth	B
58	Edge detection helps in:	Noise addition	Boundary detection	Compression	Enhancement	B
59	Preprocessing is done before:	Storage	Feature extraction	Display	Compression	B
60	Binary image contains:	Two intensity levels	Three levels	Multiple levels	Continuous values	A
61	Edge detection detects:	Color	Intensity changes	Shape	Texture	B
62	Sobel operator is used for:	Smoothing	Edge detection	Compression	Segmentation	B
63	Gradient magnitude indicates:	Color	Edge strength	Size	Shape	B
64	Corner detection identifies:	Flat regions	High variation points	Noise	Smooth areas	B
65	Harris corner detection is used for:	Noise removal	Feature detection	Compression	Enhancement	B
66	Edges occur at:	Constant intensity	Intensity discontinuity	Low variation	Smooth regions	B
67	Corner is intersection of:	Lines	Pixels	Edges	Colors	C
68	Edge detection is important for:	Compression	Object boundaries	Storage	Display	B
69	Prewitt operator is used for:	Filtering	Edge detection	Segmentation	Enhancement	B
70	Gradient is maximum at:	Flat region	Edge	Corner	Noise	B
71	Feature extraction reduces:	Noise	Data dimension	Color	Brightness	B
72	Hough Transform is	Noise removal	Shape detection	Compression	Enhancement	B

Q.No	Question	Option A	Option B	Option C	Option D	Ans
	used for:					
73	Hough Transform detects:	Edges	Lines and shapes	Noise	Colors	B
74	Feature matching compares:	Pixels	Features	Colors	Sizes	B
75	SIFT is used for:	Filtering	Feature detection	Compression	Thresholding	B
76	Scale-invariant means:	Same size	Works at different scales	Fixed scale	No scaling	B
77	Feature descriptors represent:	Pixels	Feature characteristics	Color	Shape	B
78	Matching is based on:	Distance	Size	Color	Shape	A
79	Keypoints are:	Pixels	Important features	Colors	Noise	B
80	Feature extraction helps in:	Compression	Recognition	Display	Storage	B
81	PCA reduces:	Noise	Dimensions	Size	Color	B
82	PCA finds:	Edges	Principal components	Pixels	Shapes	B
83	PCA is based on:	Mean	Variance	Color	Shape	B
84	Eigenvectors represent:	Data	Directions	Pixels	Noise	B
85	Eigenvalues represent:	Size	Importance	Color	Shape	B
86	PCA removes:	Noise	Redundancy	Color	Shape	B
87	PCA is used for:	Compression	Classification	Enhancement	Filtering	A
88	PCA transforms data into:	Same space	New space	Pixel space	Color space	B
89	PCA components are:	Independent	Correlated	Random	Fixed	A
90	PCA maximizes:	Mean	Variance	Noise	Color	B
91	Feature matching is used in:	Compression	Object recognition	Storage	Enhancement	B
92	Matching uses:	Distance metrics	Color	Size	Shape	A
93	Euclidean distance is used for:	Matching	Filtering	Compression	Segmentation	A
94	Image	Display	Analysis	Storage	Compression	B

Q.No	Question	Option A	Option B	Option C	Option D	Ans
	representation helps in:					
95	Feature matching is robust to:	Noise	Scale changes	Color	Shape	B
96	PCA helps in:	Feature reduction	Noise addition	Enhancement	Segmentation	A
97	Hough Transform is parameter-based:	True	False	Partial	None	A
98	Corner detection is important for:	Compression	Matching	Display	Storage	B
99	Feature detection helps in:	Storage	Recognition	Compression	Display	B
100	Image preprocessing improves:	Storage	Analysis accuracy	Display	Compression	B

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)

Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions